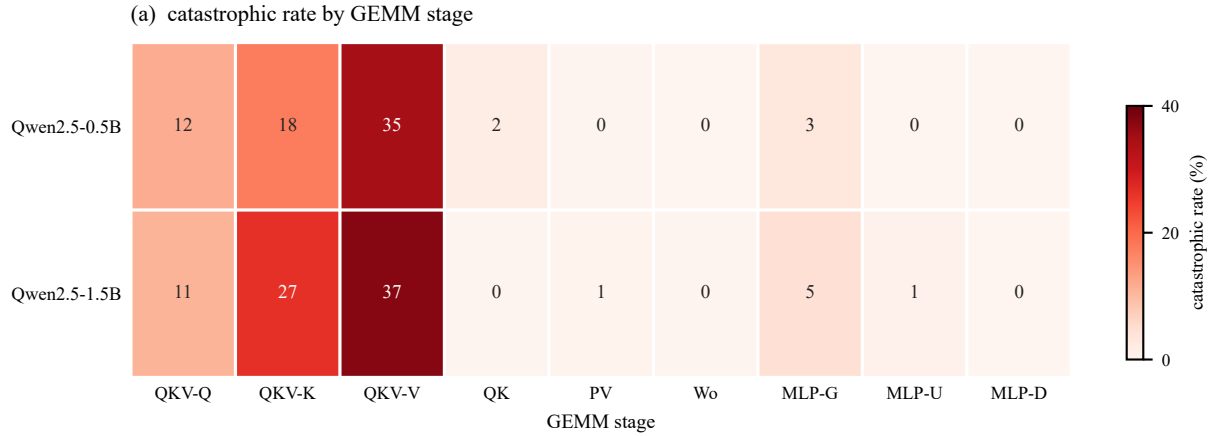
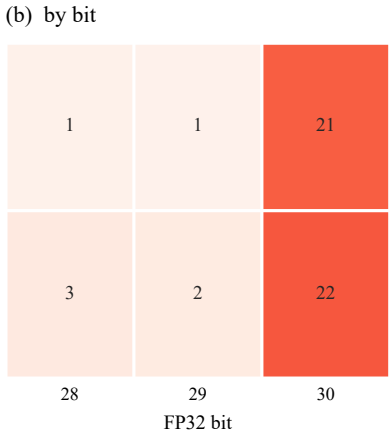
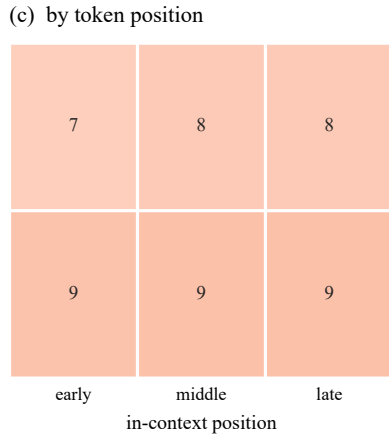
# Software GEMM bit injection in long-context agent prefill

Qwen2.5-Instruct; 12 cases; 5,832 fault runs; 648 sham; clean 24/24




one FP32 GEMM-output element flipped per run; bits 28–30; sham 648/648; all faults exactly 1 bit; four A100 shards.